EDMI

Mobile App
for Dental Microscopes

CATALOG · CONFIGURATOR · AI ASSISTANT · AR VISUALIZATION

v0.1.0 · 2026

About the Project

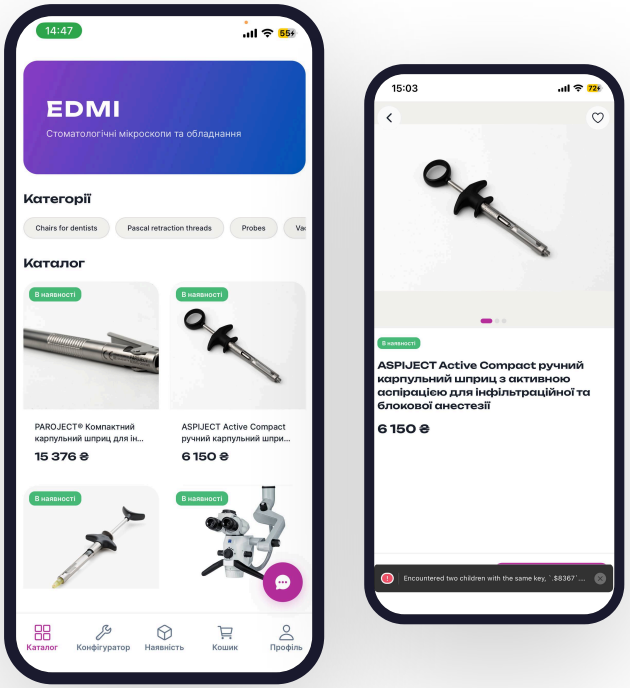
A full-featured mobile application for EDMI — a leading supplier of dental microscopes and optical equipment in Ukraine and Europe.

Catalog
Real-time WooCommerce synchronization

Configurator
Interactive microscope assembly

AI Assistant
EDMik chatbot on every screen

AR Preview
Preview microscope in your office



Product Catalog

- Marketplace-style home screen
- Hero section with gradient & search
- Quick Action buttons
- Category grid with real images
- Brand filtering & sorting
- Product cards with prices & discounts

PRODUCT DETAIL PAGE

- Photo gallery with swipe
- Badges: new / used / discount
- Specs in expandable accordions
- B2B "Request Invoice"
- Related products

Microscope Configurator

A unique feature — interactive microscope assembly tailored to client needs

- 4 models: Zeiss, Leica, CJ-Optik
- 5 option groups: color, mount, lighting, objective, camera
- Real price deltas in EUR
- Real-time cost calculation
- Add to cart with one tap

EDMik — AI Assistant

Built-in chatbot on every screen

- Floating Action Button
- 73 keywords → intent recognition
- 8 response types
- Quick Actions in chat
- Prepared for Claude AI

AR "View in Your Office"

- Full-screen AR mode
- Surface detection
- 3D model with one tap
- Gestures: drag, rotate, scale
- Fallback for non-AR devices

Cart & Checkout

Contact → Delivery → Payment → Confirm

- LiqPay + WayForPay
- Nova Poshta delivery
- Profile & order history
- Language switch UK/EN

Technical Architecture

MOBILE	React Native + Expo SDK 54
NAVIGATION	expo-router v6
STATE	Zustand v5 + TanStack Query v5
UI	Custom design system (37 components)
3D / AR	ViroReact
BACKEND	Node.js + Express + PostgreSQL + Redis
API	WooCommerce Store API
AI	Anthropic Claude API
TYPING	TypeScript (strict mode)

ARCHITECTURE

```
edmi/
├── apps/
│   ├── mobile/     ← Expo SDK 54
│   │   ├── app/    ← 18 routes
│   │   ├── components/
│   │   ├── stores/ ← 7 Zustand
│   │   └── screens/← Lazy AR
│   └── admin/      ← Vite + React
├── packages/
│   ├── shared/     ← 24 modules
│   └── server/     ← Express
└── docs/
```

→ Monorepo (pnpm workspaces)

→ WooCommerce as source of truth

→ Shared package: types + validators + i18n

→ Offline-capable catalog

→ Lazy loading for AR

Project in Numbers

18 Screens	**37** Components	**7** Stores
24 Shared modules	**2** UI Languages	**11** Bugs fixed
8 Dev phases	**6** Documents	**30MB** 3D model

Development Cost

Outsourcing Agency

$80,000 — $120,000

⏱️ 3-5 months

👥 4-5 people

📊 1,040 — 1,490 hours

VS

How It Was Actually Built

$0

⏱️ ~4 days

👤 1 person + Claude AI

🚀 AI-driven development

Savings: $80,000 — $120,000

AI-Driven Development

The architect defines the vision, AI handles the implementation

01
Planning
Phases, priorities, architecture

02
Research
AI agents analyze the codebase

03
Design
Plan approved before coding

04
Implementation
Phase-by-phase with testing

05
Debugging
Agents with evidence base

- Architect: what + why. AI: how
- Parallel research by 3 agents
- Strict TypeScript, zero tech debt
- Documentation during development

EDMI Mobile App

What's Ready ✓

- ✓ Catalog with WooCommerce data
- ✓ Detailed product pages
- ✓ Cart & 4-step checkout
- ✓ Microscope configurator
- ✓ EDMik AI chatbot
- ✓ AR visualization
- ✓ Profile, orders, multilingual

Next Steps →

- ○ Claude AI integration
- ○ Voice input (Whisper)
- ○ LiqPay / WayForPay payments
- ○ Push notifications
- ○ App Store + Google Play

Built in 4 days · $0 budget · Powered by Claude AI

edmi.com.ua